• ENGINE—4.0L OHV—CARBON KNOCK ON ACCELERATION • NOISE—CARBON KNOCK ON ACCELERATION—VEHICLES EQUIPPED WITH 4.0L OHV ENGINE ONLY	Article No. 01-19-7

FORD: 1990-1997 AEROSTAR
1990-2000 EXPLORER, RANGER

ISSUE

Some vehicles equipped with the 4.0L OHV engine may exhibit an engine noise which may be perceived by the customer as a piston/connecting rod bearing knock. This carbon knock is heard only under load during the drive cycle. Carbon knock is a customer drive duty-cycle phenomenon that cannot be repaired with an engine exchange. This may be caused by carbon build-up within the combustion chamber.

ACTION

Verify condition. If normal diagnostics fail to correct the condition, de-carbon the combustion chamber to help quiet the carbon knock noise. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

1. Use Motorcraft Carburetor Tune-up Cleaner PM-3.
2. Carbon removal:
   a. Disconnect canister purge line from throttle body.
   b. Attach a vacuum line to the canister purge port.
   c. At hot engine idle, allow the engine to ingest 1/2 to 2/3 of a can of Motorcraft Carburetor Tune-up Cleaner. Use caution not to ingest too quickly due to potential hydro-lock issues.
   d. Shut engine off and allow it to soak for one hour.
   e. Start engine, allow engine to ingest the remainder of the Motorcraft Carburetor Tune-up Cleaner.
   f. Remove vacuum line and re-install canister purge line.
   g. Road test vehicle at 3500 RPM for 2-3 miles.
   h. Repeat above procedure two times for a total of three times.
   i. Change oil and filter.
3. Review the customers' driving habits. The recommended drive cycle should include daily periods of engine operation above 3,000 RPM, such as a brisk acceleration from a stop position. This will break/burn the carbon off the piston head.
4. Use regular unleaded fuel, 87 Octane. Mid-grade and Premium fuels may increase the probability of carbon buildup, leading to a knock noise.

PART NUMBER	PART NAME
PM-3	Motorcraft Carburetor Tune-up Cleaner

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
011907A	Remove Carbon From Combustion Chamber (Includes Time To Perform Procedure A Total Of Three Times)	1.6 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
6108	42

OASIS CODES: 497000, 615000, 702000, 702100

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